POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year: 2013
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Do you consider a mobile phone (smartphone) to be intelligent? Justify your answer. 8
   b) What is 8-puzzle problem? How can you solve a problem using production system? 7
2. a) Trace the constraint satisfaction procedure for solving the following cryptarithmatic problem. 7
   SEND+MORE=MONEY
   b) Compare Depth limited search and breadth first search in terms of time and space complexity, completeness and optimality. 8
3. a) Using a suitable example, illustrate steps of A* search. Why A* search is better than Best first search. 7
   b) What are the properties possessed by a good knowledge representation systems? 8
   Give P and Q, prove that
   $((P \rightarrow Q) \rightarrow P) \rightarrow P$ is tautologous.
4. a) Convert $A \leftrightarrow B \leftrightarrow C$ into Conjunctive Normal Form. 7
   b) Represent the following sentences in first order logic: 8
      i. A person with a dust allergy sneezes.
      ii. Every flower likes water.
      iii. You can fool all of the people some of the time.
      iv. No cake lover throws a cake.
5. a) Contrast and compare Semantic Nets and Frames for knowledge representation. List limitations of these representations. 8
   b) State and explain in brief about EBL-system. What are its advantages. 7

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6. a) Define a perceptron. Describe a mechanism to train a perceptron.
   b) Describe with examples how understanding a natural language is difficult at syntactic and semantic levels.
7. Write short notes on: (Any two)
   a) Informed search Vs Uninformed search.
   b) Learning by Analogy.
   c) Baye's theorem.

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